

The optimization process used here is the same process utilized by investors to determine their optimal preferred efficient portfolio. In this depiction, the maximum level of achievable utility is U_3 and corresponds to trade strategy x . Therefore, investors with preferred portfolio A1 need to implement their investment decision utilizing strategy x to ensure consistency between investment and trading goals. Notice that in the case of a VWAP trading strategy y investors incur too much risk, resulting in a lower utility than associated with strategy x . Also, there additionally corresponds an aggressive strategy z with an equivalent level of utility as associated with the VWAP strategy. The last important point here is that there is a single “optimal” trading strategy corresponding to each efficient portfolio.

The importance of this representation in [Figure 10.5](#) is that it clearly illustrates there is a “single” optimal trading strategy consistent with the underlying investment portfolio. This is the strategy that maximizes investor utility.

COST-ADJUSTED FRONTIER

The cost-adjusted frontier is the efficient investment frontier after adjusting for trading costs (e.g., the ex-post frontier). An example of the derivation of the cost-adjusted frontier is as follows:

First, start with three efficient portfolios on the efficient investment frontier. These portfolios represent the Markowitz (ex-ante) efficient portfolios.

Second, perform trade schedule optimization for each portfolio. This will result in the set of all optimal trading strategies for each of the portfolios. It provides three different efficient trading frontiers.

Third, overlay the efficient trading frontier for each portfolio onto the efficient investment frontier. These adjusted portfolios portray the set of risk-return profiles for each of the efficient portfolios after adjusting for trading costs.

The cost-adjusted frontier is then the highest envelope of all cost-adjusted portfolios. This process is illustrated in [Figure 10.6](#) and shows multiple cost-adjusted frontiers. For each portfolio on the frontier there is a corresponding ETF. We can draw the cost-adjusted frontier as the curve through all corresponding points of the same strategy. For example, the VWAP cost-adjusted frontier is drawn by connecting all VWAP strategies on the ETF, and the same process is carried out for the aggressive